

Explain and check long multiplication

Use the scaffold, show each step and ask for an adult check when marked.

Name: _____ Date: _____

1. Read the model: Miro calculates 326×45 as 1,467. Identify the likely place-value error. Copy or complete the first step.

2. Which is closer to $4,208 \times 29$: 120,000 or 1,200,000?

3. Calculate and check $6,014 \times 38$.

4. Miro says: Miro trusts a written answer without estimating first. Is this correct? Explain.

Teacher answers and checking prompts

1. Read the model: Miro calculates 326×45 as 1,467. Identify the likely place-value error. Copy or complete the first step.

Answer He treated 40 as 4. The correct product is 14,670.

2. Which is closer to $4,208 \times 29$: 120,000 or 1,200,000?

Answer 120,000.

3. Calculate and check $6,014 \times 38$.

Answer 228,532.

4. Miro says: Miro trusts a written answer without estimating first. Is this correct? Explain.

Answer An estimate reveals place-value errors before the exact calculation is accepted.

Explain and check long multiplication

Work independently. Show a complete method and check at least one answer.

Name: _____ Date: _____

1. Miro calculates 326×45 as 1,467. Identify the likely place-value error.

6. Check this result using an inverse, estimate or known fact: Calculate and check $6,014 \times 38$.

2. Which is closer to $4,208 \times 29$: 120,000 or 1,200,000?

7. Create a four-digit by two-digit multiplication with a product between 200,000 and 210,000.

3. Calculate and check $6,014 \times 38$.

8. Identify and correct this misconception: Miro trusts a written answer without estimating first.

4. Solve this independently and show every stage: Miro calculates 326×45 as 1,467. Identify the likely place-value error.

9. Write a complete sentence explaining how you checked one answer.

5. Use a second method or representation for: Which is closer to $4,208 \times 29$: 120,000 or 1,200,000?

10. Write and solve a similar question to: Calculate and check $6,014 \times 38$.

Teacher answers and checking prompts

1. Miro calculates 326×45 as 1,467. Identify the likely place-value error.
Answer He treated 40 as 4. The correct product is 14,670.
2. Which is closer to $4,208 \times 29$: 120,000 or 1,200,000?
Answer 120,000.
3. Calculate and check $6,014 \times 38$.
Answer 228,532.
4. Solve this independently and show every stage: Miro calculates 326×45 as 1,467. Identify the likely place-value error.
Answer He treated 40 as 4. The correct product is 14,670.
5. Use a second method or representation for: Which is closer to $4,208 \times 29$: 120,000 or 1,200,000?
Answer Expected result: 120,000. Method or representation may vary.
6. Check this result using an inverse, estimate or known fact: Calculate and check $6,014 \times 38$.
Answer Expected result: 228,532. A valid check must be shown.
7. Create a four-digit by two-digit multiplication with a product between 200,000 and 210,000.
Answer Answers vary, for example $5,000 \times 41 = 205,000$.
8. Identify and correct this misconception: Miro trusts a written answer without estimating first.
Answer An estimate reveals place-value errors before the exact calculation is accepted.
9. Write a complete sentence explaining how you checked one answer.
Answer Answer should name an inverse, estimate, boundary, known fact or equivalent representation.
10. Write and solve a similar question to: Calculate and check $6,014 \times 38$.
Answer Answers vary. The new question must use the same mathematical structure and include a correct solution.

Explain and check long multiplication

Prove, compare or generalise. Write enough reasoning for another pupil to follow.

Name: _____ Date: _____

1. Create a four-digit by two-digit multiplication with a product between 200,000 and 210,000.

6. Create a new example that tests the same idea as: Calculate and check $6,014 \times 38$.

2. Prove the answer to this question, rather than only calculating it: Miro calculates 326×45 as 1,467. Identify the likely place-value error.

7. Find a second method or representation. Compare its efficiency with your first method.

3. Solve this in two different ways and compare them: Which is closer to $4,208 \times 29$: 120,000 or 1,200,000?

8. Write one always, sometimes or never statement about today's learning and justify it.

4. Work backwards from the answer to reconstruct the question: 228,532.

9. Create a believable incorrect solution. Identify the exact step where the reasoning breaks.

5. Explain why this reasoning fails: Miro trusts a written answer without estimating first.

10. Change one value or condition in a question. Explain what changes in the solution and what stays the same.

Teacher answers and checking prompts

1. Create a four-digit by two-digit multiplication with a product between 200,000 and 210,000.
Answer Answers vary, for example $5,000 \times 41 = 205,000$.
2. Prove the answer to this question, rather than only calculating it: Miro calculates 326×45 as 1,467. Identify the likely place-value error.
Answer Expected result: He treated 40 as 4. The correct product is 14,670. Proof or justification must match the lesson structure.
3. Solve this in two different ways and compare them: Which is closer to $4,208 \times 29$: 120,000 or 1,200,000?
Answer Expected result: 120,000. Two valid methods and a comparison are required.
4. Work backwards from the answer to reconstruct the question: 228,532.
Answer One valid reconstruction is: Calculate and check $6,014 \times 38$. Other equivalent questions are acceptable.
5. Explain why this reasoning fails: Miro trusts a written answer without estimating first.
Answer An estimate reveals place-value errors before the exact calculation is accepted.
6. Create a new example that tests the same idea as: Calculate and check $6,014 \times 38$.
Answer Answers vary. The example and solution must preserve the mathematical structure.
7. Find a second method or representation. Compare its efficiency with your first method.
Answer Answers vary. Comparison should refer to accuracy, number structure and clarity.
8. Write one always, sometimes or never statement about today's learning and justify it.
Answer Answers vary. A valid example and counterexample should support the classification.
9. Create a believable incorrect solution. Identify the exact step where the reasoning breaks.
Answer Answers vary. The error must be mathematically relevant and the correction must be precise.
10. Change one value or condition in a question. Explain what changes in the solution and what stays the same.
Answer Answers vary. The explanation must distinguish the mathematical structure from the changed value.